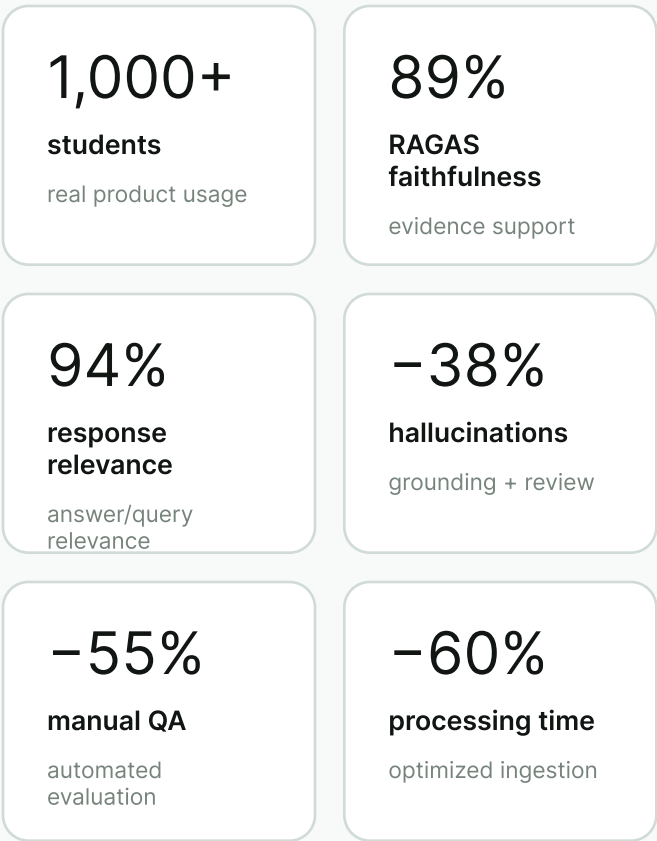


ThirdSlate

Agentic RAG for Course-Grounded Learning

Course-scoped RAG • Grounding • Evaluation • Human Review • Kubernetes

Co-founder & AI Engineer



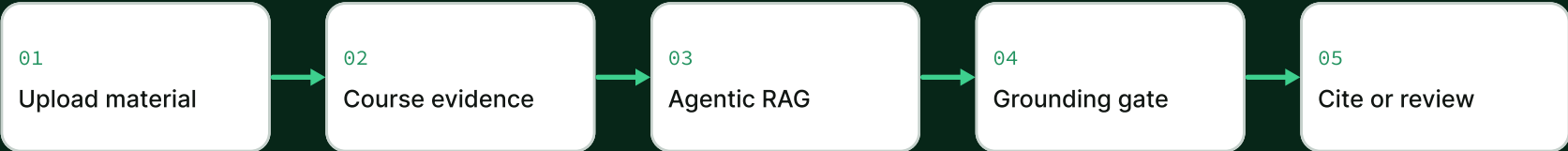
ENGINEERING NARRATIVE

Course material → scoped retrieval → agent orchestration → grounding → evidence-backed response

I co-founded ThirdSlate and personally designed the UI/UX plus the AI/backend path from ingestion and retrieval through generation, evaluation, persistence, authentication, and deployment.

STUDENT PRODUCT LOOP

From uploaded course material to a grounded, reviewable answer.



ENGINEERING SCOPE

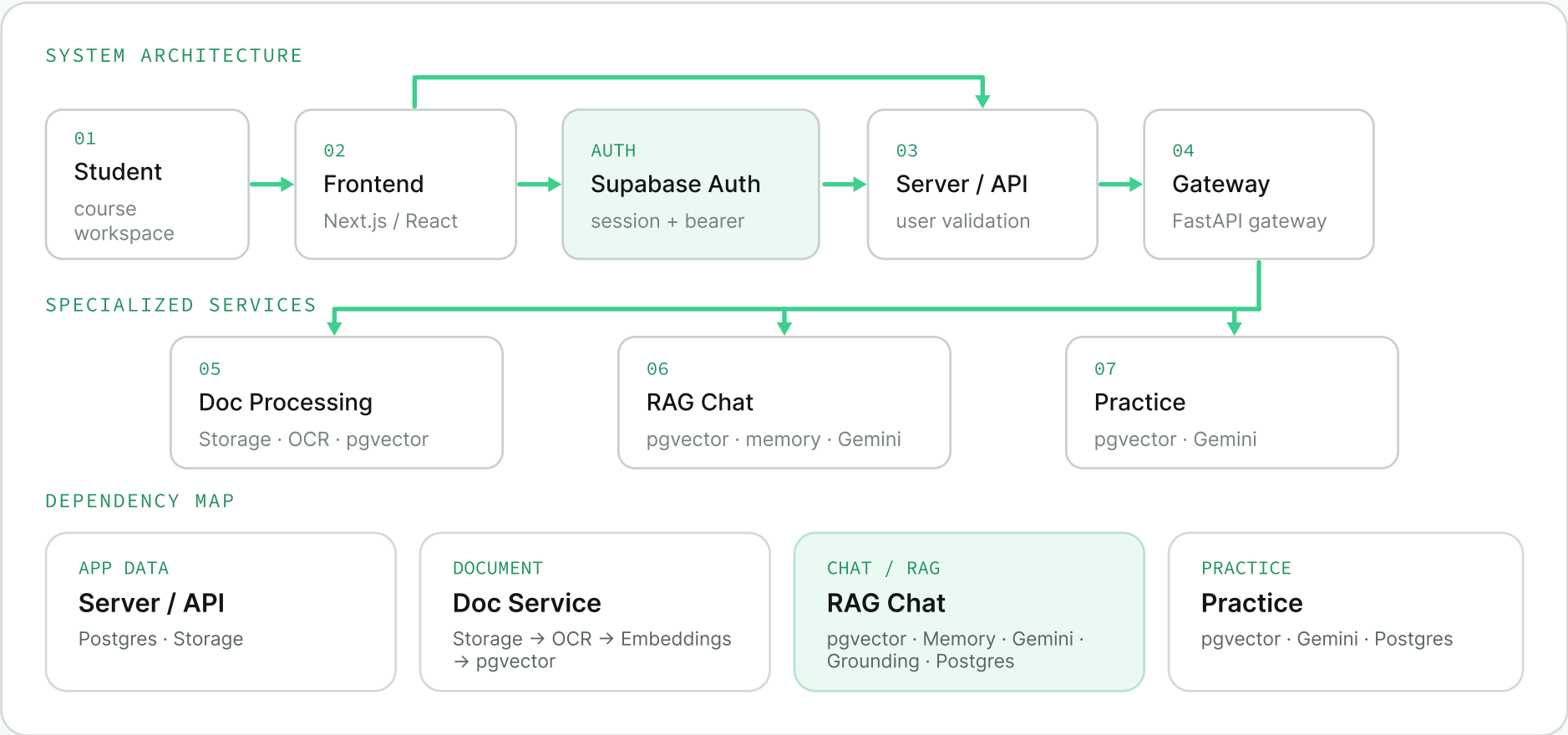
- Agentic RAG
- Grounding
- Evaluation
- Human Review
- Document AI
- AWS EKS

CORE TECHNOLOGIES

LangGraph • Gemini • OpenAI embeddings • pgvector • Mem0 • RAGAS • DeepEval
FastAPI • Supabase • PyMuPDF • OCR • Docker • Kubernetes • AWS EKS • GitHub Actions

Product Problem, Ownership & System Architecture

How the student experience, identity boundary, AI services, and data systems fit together.



OWNERSHIP LEDGER

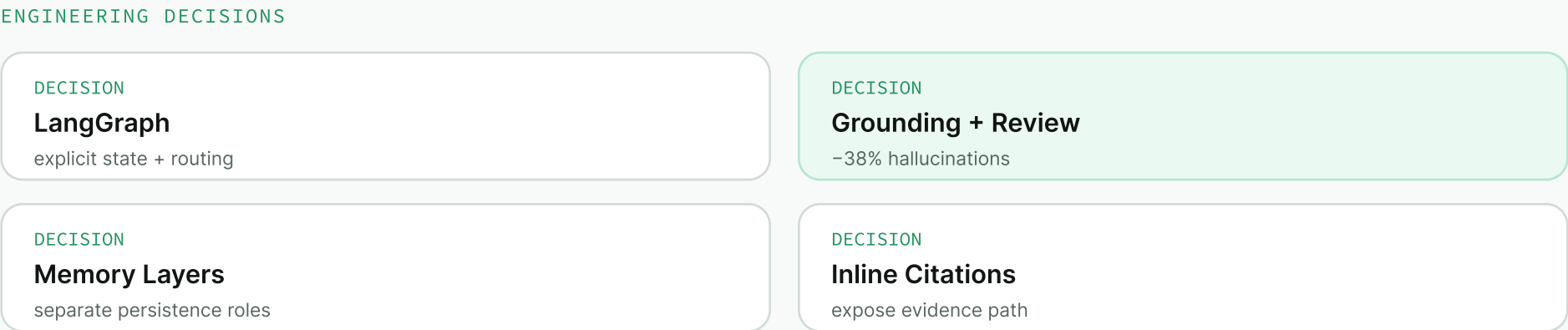
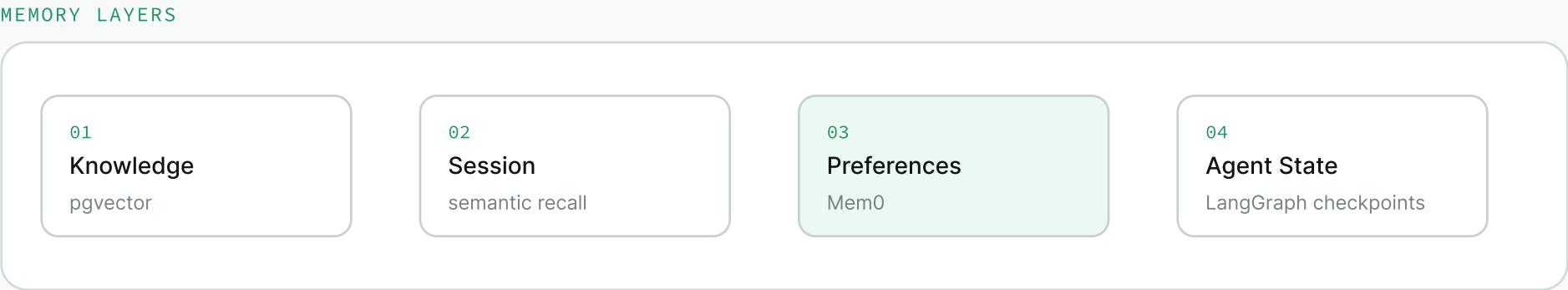
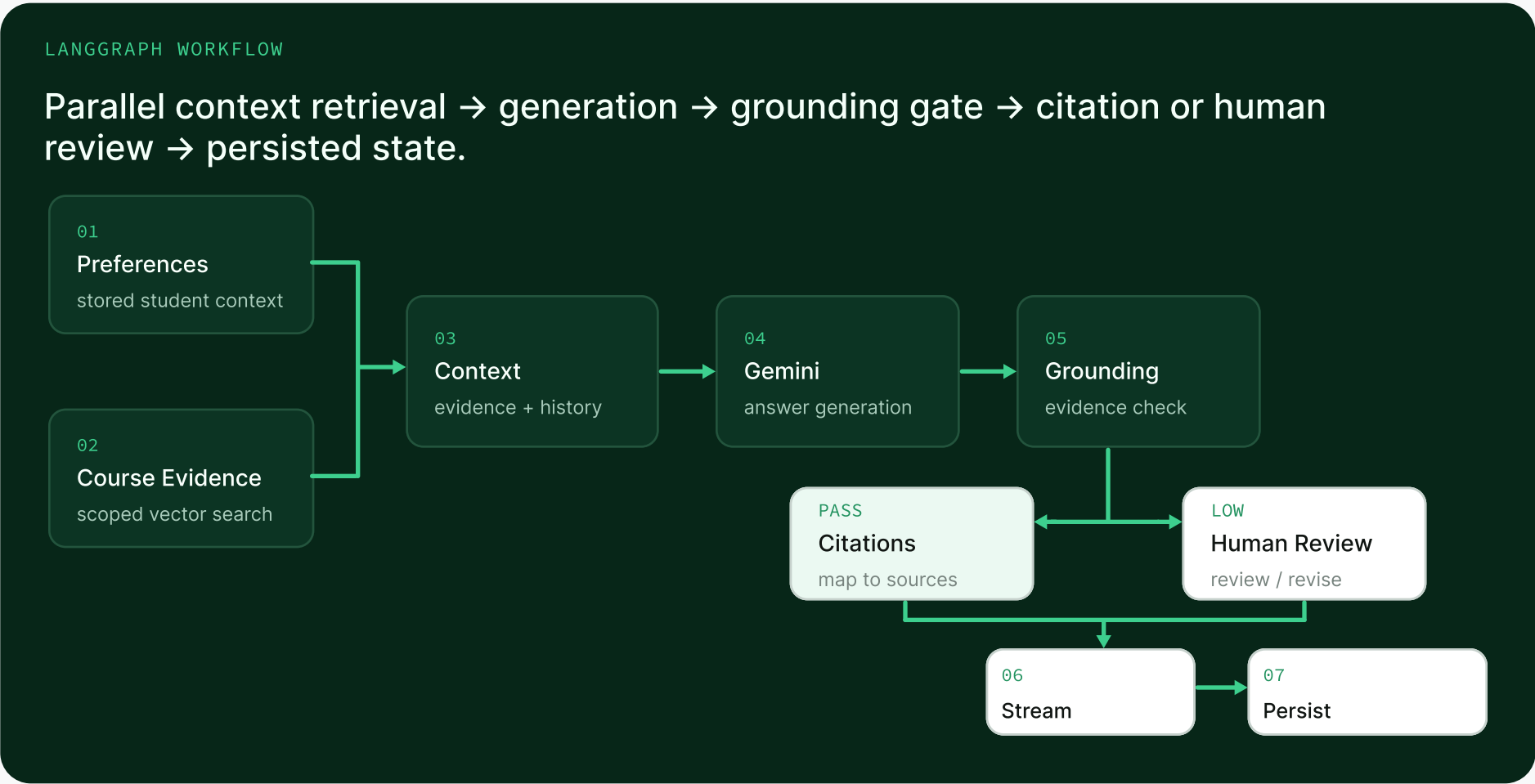
PRIMARY OWNER	UI / UX complete UI / UX	PRIMARY OWNER	Agent Backend LangGraph + RAG
PRIMARY OWNER	Grounding critic + human review	PRIMARY OWNER	Evaluation 500+ cases / 8 domains
PRIMARY OWNER	Ingestion PDF + DOCX + OCR	PRIMARY OWNER	Auth Boundary Supabase + JWT
PRIMARY OWNER	Domain Services 3 FastAPI services	PRIMARY OWNER	Deployment Docker + EKS + CI/CD

PROBLEMS SOLVED

SYSTEM GOAL Course Grounding course material as evidence	SYSTEM GOAL Data Isolation user · course · material scope
SYSTEM GOAL Traceable Answers inline source citations	SYSTEM GOAL Deployable AI containerized Kubernetes path

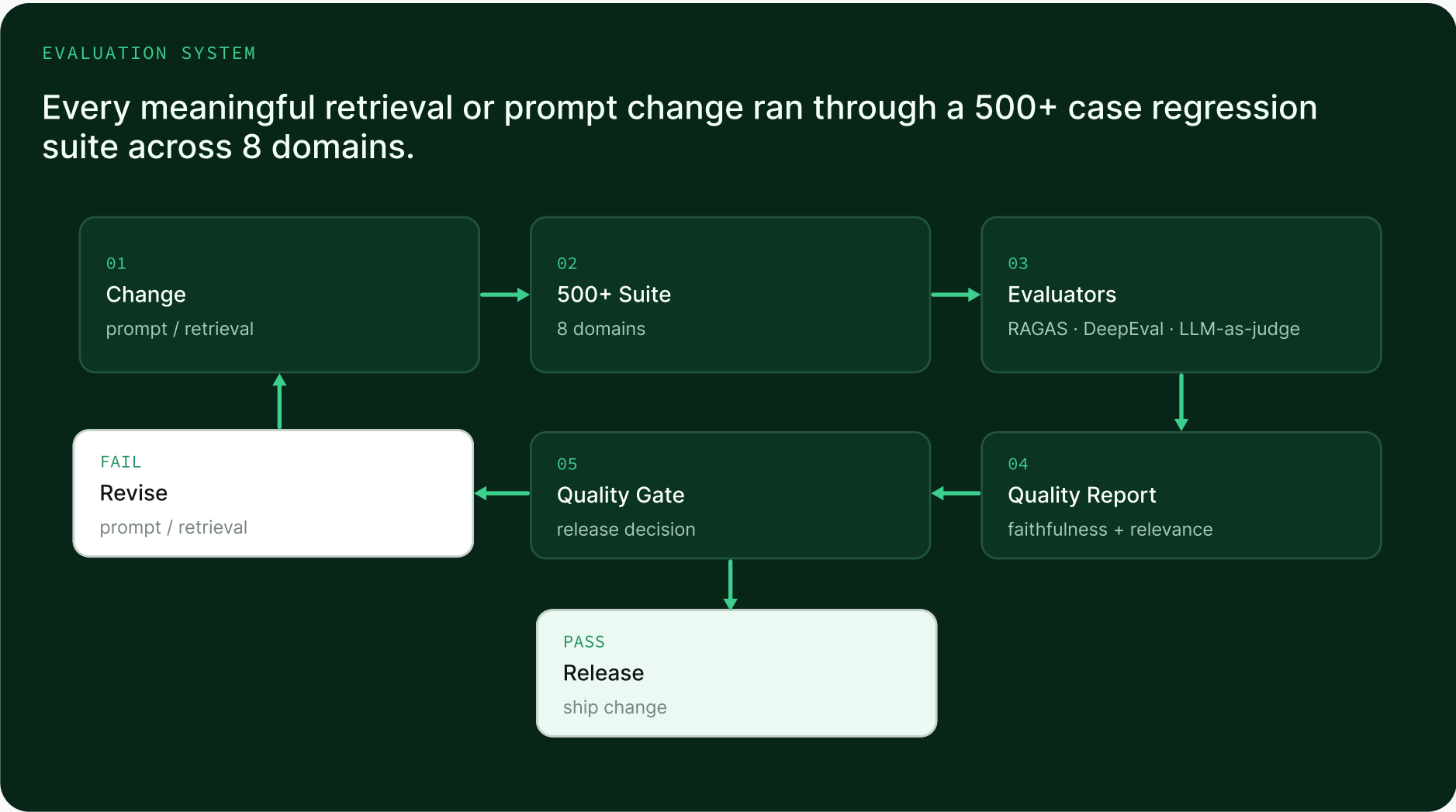
Agentic RAG, Grounding & Memory

An explicit state machine for retrieval, generation, grounding, citations, human review, and resumable persistence.

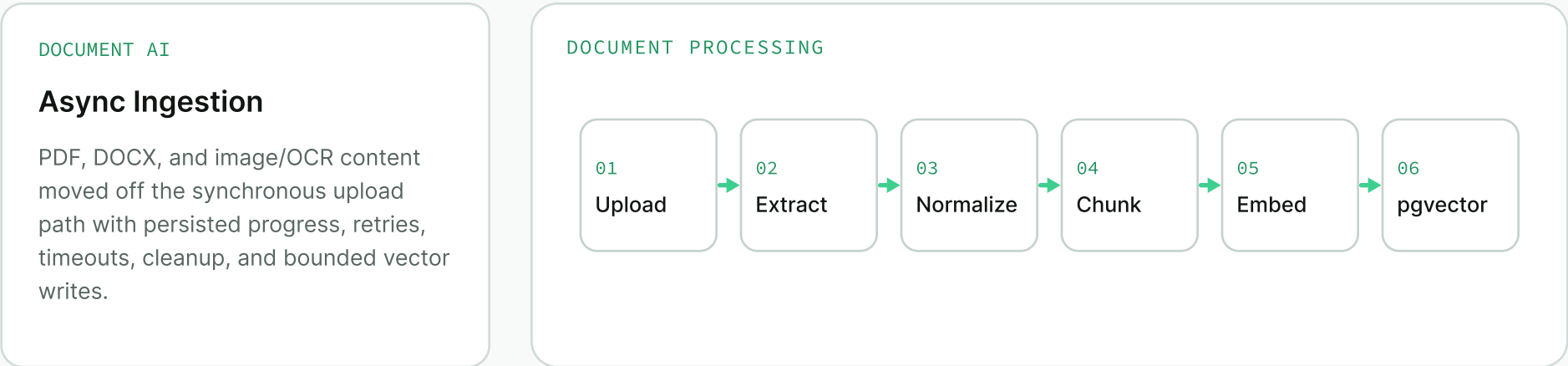


Evaluation & Ingestion Engineering

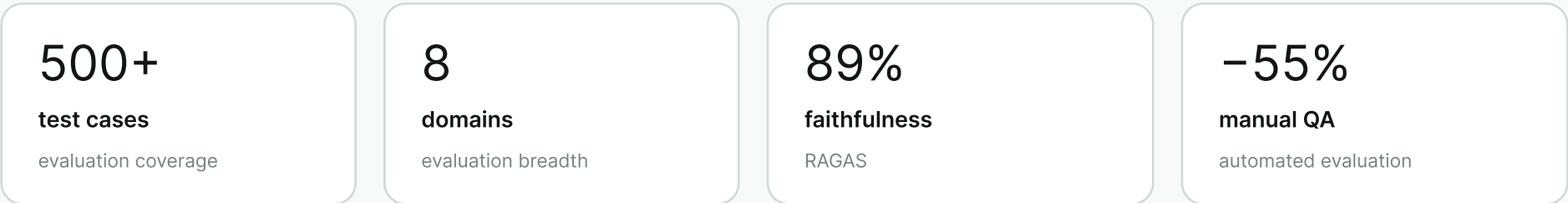
Quality gates made retrieval and prompt changes repeatable; asynchronous multimodal ingestion turned student files into retrieval-ready evidence.



INGESTION PIPELINE



QUALITY SIGNALS



Ingestion outcome · 60% lower document-processing time after optimizing multimodal extraction, OCR, chunking, embedding, and vector ingestion.

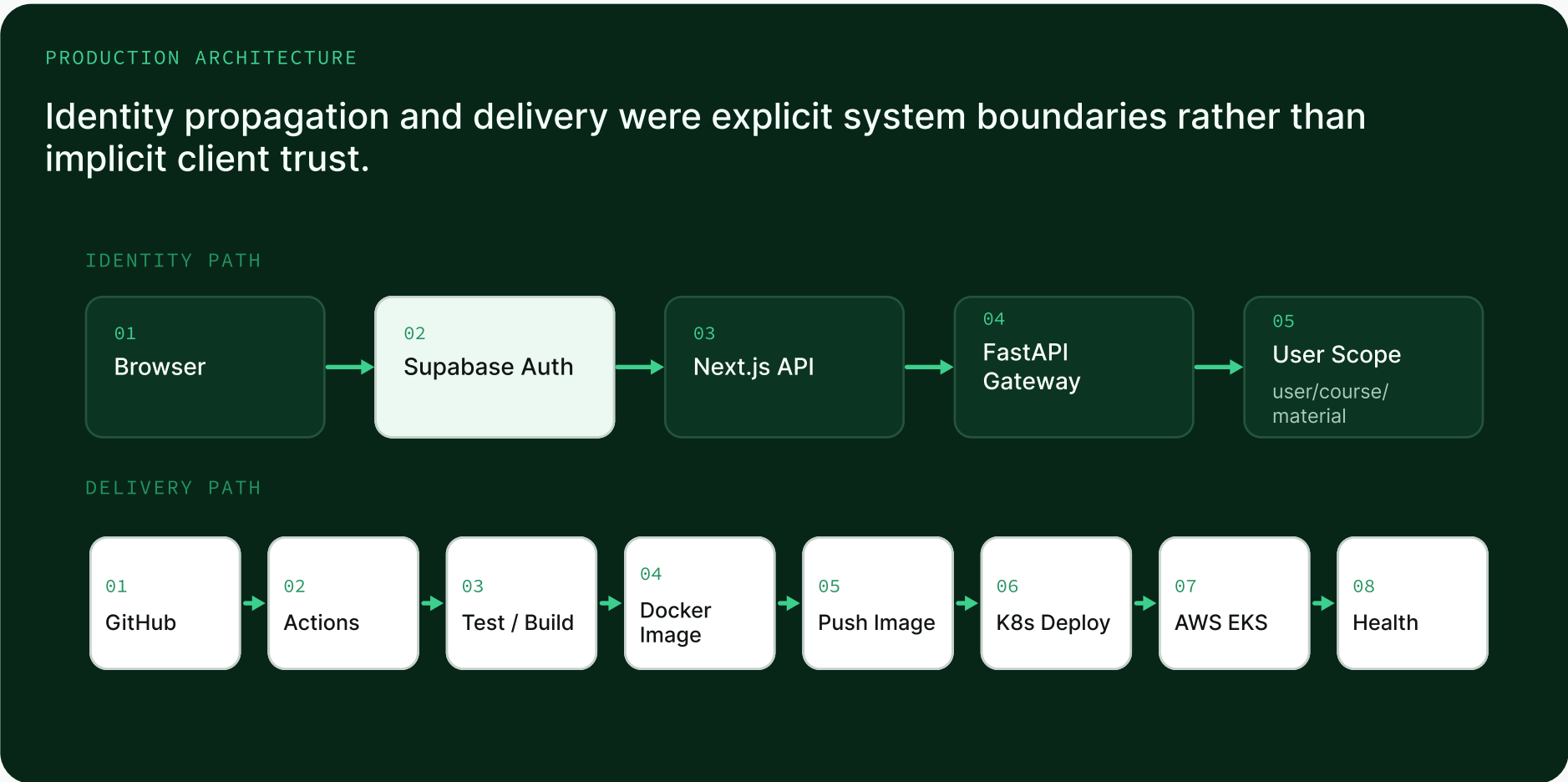
Microservices, Security & Deployment

Three domain FastAPI services behind an authenticated FastAPI gateway, deployed to AWS EKS.

SYSTEM BOUNDARY

3 Services + Gateway

Document processing, RAG chat, and practice workflows stayed independently scoped while authenticated identity propagated from Supabase through the server/API and FastAPI layers.



PRODUCTION CONTROLS

SECURITY

Scoped Identity

Server-side auth and JWT propagation scoped downstream retrieval to authenticated students.

ARCHITECTURE

Service Boundaries

Ingestion, RAG, and practice workflows had distinct responsibilities and failure modes.

RELIABILITY

Runtime Controls

Streaming, async processing, retries, timeouts, bounded writes, checkpoints, health checks, and structured logs.

PRODUCTION DELIVERY

AWS EKS

Python + FastAPI services were containerized with Docker and released through a GitHub Actions CI/CD pipeline. Quality tooling included Pytest, Pydantic, Black, Ruff, and health/readiness checks.

Results, Engineering Decisions & Lessons

A compact audit trail of the product outcomes, architecture choices, and reliability lessons from building ThirdSlate end to end.

CONSOLIDATED RESULTS

SYSTEM	RESULT	FRAMING
Product	1,000+ students	Real product usage
AI quality	89% faithfulness 94% relevance	500+ cases / 8 domains
Grounding / QA	–38% hallucinations –55% manual QA	Prior baseline + automated QA
Ingestion	–60% processing time	Optimized multimodal pipeline

ARCHITECTURE DECISIONS

ADR LangGraph Use explicit state and routing for retrieval, grounding, review, citations, and resumability.	ADR Service Boundaries Separate ingestion, chat, and practice because their workloads and failure modes differ.
ADR Memory Layers Keep knowledge, session recall, preferences, and graph state as distinct persistence concerns.	ADR Eval Gates Gate prompt and retrieval changes on repeatable evaluation instead of subjective spot checks.

ENGINEERING LESSONS

01 Grounding is Control Reliability improved when grounding became an explicit decision path, not only a prompting instruction.	02 Eval Before Release A maintained benchmark made retrieval and prompt changes comparable and reduced repetitive manual QA.	03 AI is Systems Useful model behavior depended on ingestion, identity, memory, routing, persistence, and deployment working together.
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PUBLIC BOUNDARY

High-level architecture

Reported metrics

No confidential data

Production-backed